UNIVARIATE ANALYSIS

Central Tendency:

	Age	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio	Selector
mean	44.746141	1.921441	0.879674	251.258576	47.893225	64.833619	6.487307	3.141852	0.941026	0.713551
median	45.0	1.0	0.3	208.0	35.0	42.0	6.6	3.1	0.947064	1.0
mode	60	0.8	0.2	481.75	116.75	180	7.0	3.0	1.0	1

From the table, The analysis of the average age of the patient is 44 by 'mean' and most of the patient in 60's by 'mode' function.

Outliers:

```
Greater
```

```
['TB', 'DB', 'Alkphos', 'Sgpt', 'Sgot', 'TP', 'A/G Ratio']
```

```
Lesser
```

```
['TP']
```

	Age	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio	Selector
lesser	-4.5	-1.9	-1.45	-8.25	-33.25	-68.0	3.7	0.8	0.1	-1.5
greater	95.5	5.3	2.95	481.75	116.75	180.0	9.3	5.6	1.7	2.5

- The outliers value of features in dataset.
- To detect by IQR and replacing the value.
- If the value in dataset is less than lesser, replace the ow outliers
- If the value is greater than greater outlier, replace the high outliers.

Replacing the outliers:

```
for column in Lesser:
    dataset[column][dataset[column]<descriptive.loc['lesser',column]]=descriptive.loc['lesser',column]
for column in Greater:
    dataset[column][dataset[column]>descriptive.loc['greater',column]]=descriptive.loc['greater',column]
```

Finding and Replacing NullValues:

```
dataset.isnull().sum()
```

```
Age          0
Gender       0
TB           0
DB           0
Alkphos      0
Sgpt         0
Sgot         0
TP           0
ALB          0
A/G Ratio    4
Selector     0
dtype: int64
```

```
dataset['A/G Ratio'].fillna(dataset['A/G Ratio'].mean(),inplace=True)
```

- Replacing the null values by mean to fill Nan values.
- The dataset should be without outliers and Nan values, if not the model couldn't show the accuracy result.

Standard Normal Distribution:

- It standardize (define range) the data for analysis by Z_score
- Here mean is always 0 and standard deviation is 1.
- Z_score is the test only for univariate analysis. It converts the normal distribution into standard normal distribution